

YASHASWI SHARMA

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AWS-certified Machine Learning Engineer

EDUCATION

University of Southern California

M.S. Computer Science

Los Angeles, CA, USA

08/2025 - 05/2027

Research: LLM Post-Training (Code, Tool-Use, Economic Strategy, Time-Series) - Melady Lab

Thesis (in progress): Post-Training LLMs via Economic Games in RLVR | **Advisor:** Yan Liu

University of Maryland

B.S. Computer Science (Machine Learning) & B.S. Economics (Macro, Game Theory)

College Park, MD, USA

08/2018 - 05/2022

PUBLICATIONS & RESEARCH

AgentX OpenEnv Competition: Reinforcement Learning Environment Suite

PyTorch / HuggingFace / TRL / Unsloth / FSDPv2 · 2026 - Lead (5 envs), Mentor (1 env) Under Review

- Proposed **economic game MDPs as proxy curricula** for RLVR post-training of generalizable reasoning primitives (adaptive experimental design, sequential commitment, Bayesian type inference) with closed-form verifiable rewards; trained end-to-end via GRPO + distributed vLLM (8×H100s)
- **ReasoningEconomicsEnv**: LLM allocates thinking tokens across 10 sequential MetaMathQA problems under a shared episode budget; trains difficulty-aware compute allocation
- **SearchEconomicsEnv**: LLM queries search API under per-call cost penalty; commit reward grounded in Weitzman/Pandora's Box optimal-stopping theory; evaluated on HotpotQA
- **ToolOrchestratorEnv** (*mentor*): cost-aware tool selection across 6 heterogeneous tools with Weitzman reservation prices
- **LotteryElicitationEnv**: LLM proposes lottery pairs to a closed-form Kahneman-Tversky respondent to recover latent (γ, λ) ; first RL environment using RLVR to train sequential Bayesian type inference with verifiable ground-truth reward
- **MiniGridEnv**: OpenEnv adapter over BabyAI/MiniGrid; post-episode LLM memory rewrite enables cross-episode knowledge retention via branch-stable GRPO rollouts
- **TemporalBenchEnv**: MCQ-style RL environment over TemporalBench

TemporalBench: A Benchmark for Contextual and Event-Informed Time-Series Forecasting

Knowledge Discovery & Data Mining 2026 - 2nd Author - Paper

Accepted

The Meta-Reasoning Gap: On the Limits of Inference-Time Cost Prediction

Conference on Language Modeling 2026 - 1st Author

Under Review

Identified limitations of LLMs' to predict reasoning difficulty, establishing foundational constraints for test-time compute allocation. Inspiration for **ReasoningEconomicsEnv**

WORK EXPERIENCE

Snovation

AI Engineer

Remote-Baltimore, MD, USA

12/2024 - 07/2025

- Engineered an application-specific LLM hosting service, maximizing token throughput by 200X (20,000 %) (from 0.34 to 67.1 tok/s) and reducing inference latency by 77% (from 180 sec to 42 sec) while optimizing resource utilization to a CPU-only architecture using 2 VCPUs

Amazon Web Services

Software Development Engineer

Arlington, VA, USA

09/2022 - 09/2024

- Designed & deployed the inference and feedback systems for AWS Config's first Generative AI feature (**ReInvent 2023**; Advanced Query Assistant)
- Developed a core data management system with 99.99% availability & performance, leveraging Day-0 AWS services for state management, message processing, and anti-entropy mechanisms
- Led 200+ critical on-call investigations across diverse systems, including those with major global security implications, ensuring 24/7 system availability, reliability, and security

SKILLS

LLM Post-Training:	TRL, Unsloth, vLLM, HuggingFace, PEFT/LoRA, FSDPv2, DeepSpeed ZeRO, Accelerate, veRL, OpenEnv, litellm, evalplus
ML Systems & Inference:	PyTorch, Tensorflow, Quantization (GPTQ/AWQ), CUDA, MPI, OpenMP, JAX/XLA, Triton, SageMaker, Inferentia
Infrastructure & Distributed Systems:	Docker, Kubernetes/EKS, EC2, gRPC/Protobuf, REST, AWS Lambda/Step Functions, Azure Serverless
Data Engineering:	Spark, DynamoDB, Kinesis, Redshift, OpenSearch, RDS, EMR, Batch, S3
Programming Languages:	Python, Go, Rust, C/C++, Java, Kotlin, OCaml, TypeScript, Javascript, SQL